CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 8: Introduction to Trigonometry

Previous year board questions, topic-wise: 27 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Trigonometric ratios

1. If $\sin\theta = \frac{a}{b}$, write $\cos\theta$ in terms of a and b. (a) $\frac{b}{\sqrt{b^2-a^2}}$ (b) $\frac{b}{a}$ (c) $\frac{\sqrt{b^2-a^2}}{b}$ (d) $\frac{a}{\sqrt{b^2-a^2}}$ CBSE 2026, 1 mark

2. If $\tan\theta = \frac{24}{7}$, find $\sin\theta + \cos\theta$. CBSE 2026, 2 marks

3. If $\cot\theta = \frac{7}{8}$, evaluate $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$. CBSE 2026, 2 marks

4. If $2\tan A = 3$, find the value of $\frac{4\sin A + 3\cos A}{4\sin A - 3\cos A}$. (a) $\frac{7}{\sqrt{13}}$ (b) $\frac{1}{\sqrt{13}}$ (c) 3 (d) does not exist CBSE 2023, 1 mark

5. If $\sin\alpha = \frac{1}{\sqrt{2}}$ and $\cot\beta = \sqrt{3}$, find $\operatorname{cosec}\alpha + \operatorname{cosec}\beta$. CBSE 2023, 2 marks

6. If $4\tan\theta = 3$, evaluate $\frac{4\sin\theta - \cos\theta + 1}{4\sin\theta + \cos\theta - 1}$. CBSE 2018, 3 marks

2 Ratios of standard angles

7. If $\cos A = \frac{1}{2}$, find the value of $\sin^2 A + 2\cos^2 A$. (a) $\frac{3}{2}$ (b) $\frac{5}{4}$ (c) -1 (d) $\frac{1}{2}$ CBSE 2026, 1 mark

8. Find the acute angle θ for which $7 + 4\sin\theta = 9$. CBSE 2025, 1 mark

9. Given that $x\cos 60^\circ + y\cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5$, find the value of $x + 2y$. CBSE 2025, 2 marks

10. Evaluate $\frac{\tan^2 60^\circ}{\sin^2 60^\circ + \cos^2 30^\circ}$. CBSE 2025, 2 marks

11. Given $\cos(\alpha+\beta) = 0$, find $\cos\left(\frac{\alpha+\beta}{2}\right)$. (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$ (c) 0 (d) $\sqrt{2}$ CBSE 2024, 1 mark

12. Find the value of $2\sqrt{2}\cos 45^\circ \sin 30^\circ + 2\sqrt{3}\cos 30^\circ$. CBSE 2024, 2 marks

13. For $A = 60^\circ$ and $B = 30^\circ$, verify that $\sin(A+B) = \sin A\cos B + \cos A\sin B$. CBSE 2024, 2 marks

14. Evaluate $\frac{3}{4}\tan^2 30^\circ - \sec^2 45^\circ + \sin^2 60^\circ$. (a) -1 (b) $\frac{5}{6}$ (c) $-\frac{3}{2}$ (d) $\frac{1}{6}$ CBSE 2023, 1 mark

3 Trigonometric identities

15. Given $x = h + a\cos\theta$ and $y = k + b\sin\theta$, prove that $\left(\frac{x-h}{a}\right)^2 + \left(\frac{y-k}{b}\right)^2 = 1$. CBSE 2026, 3 marks

16. Prove that $\dfrac{\tan A}{1+\sec A} - \dfrac{\tan A}{1-\sec A} = 2\,\text{cosec}\,A$. CBSE 2026, 3 marks

17. Simplify $\tan^2\theta - \left(\dfrac{1}{\cos\theta} \times \sec\theta\right)$. CBSE 2025, 1 mark

18. Prove that $\dfrac{\tan\theta}{1-\cot\theta} + \dfrac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\text{cosec}\,\theta$. CBSE 2025, 3 marks

19. Prove that $\dfrac{\sin A+\cos A}{\sin A-\cos A} + \dfrac{\sin A-\cos A}{\sin A+\cos A} = \dfrac{2}{2\sin^2 A - 1}$. CBSE 2025, 3 marks

20. Given $\sec\theta - \tan\theta = m$, find $\sec\theta + \tan\theta$. (a) $1 - \dfrac{1}{m}$ (b) $m^2 - 1$ (c) $\dfrac{1}{m}$ (d) $-m$ CBSE 2024, 1 mark

21. Prove the identity $\dfrac{\tan\theta}{1-\cot\theta} + \dfrac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\text{cosec}\,\theta$. CBSE 2024, 3 marks

22. If $\sin\theta + \cos\theta = \sqrt{3}$, find the value of $\sin\theta\cos\theta$. CBSE 2023, 2 marks

23. Prove that $\dfrac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} = \dfrac{1+\sin\theta}{\cos\theta}$. CBSE 2023, 3 marks

24. Prove that $(\sin\theta + \text{cosec}\,\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$. CBSE 2019, 3 marks

25. Prove that $(1 + \cot A - \text{cosec}\,A)(1 + \tan A + \sec A) = 2$. CBSE 2019, 3 marks

26. Prove that $\dfrac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = \dfrac{1}{\sec A - \tan A}$. CBSE 2019, 4 marks

27. Prove that $\dfrac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A$. CBSE 2018, 4 marks

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